

Laplacian Connectivity Force for Scaffold Inpainting

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July 1, 2026

1 Introduction

In scaffold inpainting, a diffusion model generates fragment atoms $\mathcal{F}$ around a fixed scaffold context $\mathcal{C}$. After reverse diffusion, a bond classifier assigns edges from geometry. A persistent failure mode is *disconnectedness*: generated fragment atoms drift far enough from $\mathcal{C}$ that no bond is classified between them and the scaffold, leaving the fragment as a separate, valueless component.

Existing drift confinement (a harmonic well pulling the fragment centre-of-mass toward the scaffold COM) reduces but does not eliminate this. We introduce a differentiable *Laplacian connectivity force* applied at every denoising step that directly maximises the algebraic connectivity of the fragment-scaffold graph.

2 Spectral Graph Theory

2.1 Laplacian spectrum

Let $G = (\mathcal{V}, \mathcal{E})$ be an undirected graph on $n = |\mathcal{V}|$ vertices. Define the adjacency matrix $A \in \mathbb{R}^{n \times n}$ and degree matrix $D = \text{diag}(d_1, \dots, d_n)$ where $d_i = \sum_j A_{ij}$. The graph Laplacian is the discrete analogue of the continuous Laplace operator: just as the Laplacian of a smooth function measures the deviation of a point's value from the local average of its neighbourhood, the graph Laplacian measures how much a vertex's value differs from the values of its neighbours. The *combinatorial graph Laplacian* is

$$L = D - A. \tag{1}$$

L is symmetric positive semi-definite for any graph. Order its eigenvalues as $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$.

2.2 Eigenvalue zero and connected components

The fundamental spectral theorem for graphs states:

Theorem 1 (Fiedler [1], 1973). *The geometric multiplicity of the eigenvalue 0 of L equals the number of connected components of G .*

Proof sketch. First we establish the quadratic-form identity. For any $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{x}^T L \mathbf{x} = \mathbf{x}^T D \mathbf{x} - \mathbf{x}^T A \mathbf{x} = \sum_i \left(\sum_j A_{ij} \right) x_i^2 - \sum_{i,j} A_{ij} x_i x_j \quad (2)$$

$$= \sum_{i,j} A_{ij} x_i^2 - \sum_{i,j} A_{ij} x_i x_j = \sum_{i,j} A_{ij} (x_i^2 - x_i x_j). \quad (3)$$

Interchanging the dummy indices $i \leftrightarrow j$ and averaging the two expressions yields:

$$\mathbf{x}^T L \mathbf{x} = \frac{1}{2} \sum_{i,j} A_{ij} (x_i - x_j)^2 \stackrel{A_{ij} \in \{0,1\}}{=} \sum_{(i,j) \in \mathcal{E}} (x_i - x_j)^2, \quad (4)$$

where the right-hand specialization is the unweighted (binary-adjacency) case used in this section: the factor $\frac{1}{2}$ cancels the double-counting of each unordered edge over the two ordered pairs (i, j) and (j, i) .

Second, since L is symmetric positive semi-definite, $\mathbf{x}^T L \mathbf{x} \geq 0$ for all $\mathbf{x}$, with equality iff $\mathbf{x} \in \ker L$ (a positive semi-definite quadratic form vanishes only on the null space of its matrix). Hence $L \mathbf{x} = 0 \iff \mathbf{x}^T L \mathbf{x} = 0$.

Thus, $\mathbf{x}^T L \mathbf{x} = 0$ iff $x_i = x_j$ for every edge (i, j) , i.e. $\mathbf{x}$ is constant on each connected component. The dimension of this null space is therefore the number of components. $\square$

Since L is real symmetric, the spectral theorem gives an orthonormal eigenbasis, so geometric and algebraic multiplicities coincide for every eigenvalue of L : the multiplicity of the eigenvalue 0 equals $\dim \ker L$, the dimension of its eigenspace, consistent with Theorem 1.

2.3 The Fiedler value

The second-smallest eigenvalue λ_2 is the *algebraic connectivity* (or *Fiedler value*) of G . By Theorem 1:

$$\lambda_2 = 0 \iff G \text{ is disconnected} \quad (5)$$

$$\lambda_2 > 0 \iff G \text{ is connected.} \quad (6)$$

Moreover, larger λ_2 indicates a “harder to disconnect” (more robustly connected) graph: it is the graph-theoretic analogue of spectral gap.

Theorem 2 (Courant–Fischer / Rayleigh characterization).

$$\lambda_2 = \min_{\substack{\mathbf{x} \perp \mathbf{1} \\ \mathbf{x} \neq 0}} \frac{\mathbf{x}^T L \mathbf{x}}{\mathbf{x}^T \mathbf{x}}. \quad (7)$$

Proof sketch. L is symmetric, so its eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ form an orthonormal basis, with $\mathbf{v}_1 = \mathbf{1}/\sqrt{n}$ the constant eigenvector for $\lambda_1 = 0$. Any $\mathbf{x} \perp \mathbf{1}$ expands as $\mathbf{x} = \sum_{k \geq 2} c_k \mathbf{v}_k$, so

$$\frac{\mathbf{x}^T L \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \frac{\sum_{k \geq 2} c_k^2 \lambda_k}{\sum_{k \geq 2} c_k^2} \geq \lambda_2, \quad (8)$$

with equality at $\mathbf{x} = \mathbf{v}_2$. This is the Courant–Fischer min-max theorem restricted to the second eigenvalue. $\square$

This variational characterization gives λ_2 a natural interpretation as a measure of connectivity. By the quadratic form identity (4), the numerator $\mathbf{x}^T L \mathbf{x} = \sum_{(i,j) \in \mathcal{E}} (x_i - x_j)^2$ is the graph’s *Dirichlet energy*: the cost of assigning values to vertices that disagree across edges. Equation (7) therefore says λ_2 is the minimum Dirichlet energy over all *non-constant* vertex assignments. By Courant–Fischer, increasing λ_2 increases the minimum energy required to make the graph look disconnected (i.e. to assign nearly constant values on each side of a near-cut). Consequently, increasing λ_2 discourages disconnected configurations. The Fiedler vector $\mathbf{v}_2$ attaining the minimum is also the classical basis for spectral graph partitioning: thresholding $\mathbf{v}_2$ yields the standard *sparsest cut* heuristic [3], so the connectivity force derived in Section 5 concentrates on the edges a spectral partitioner would identify as weakest.

3 A Differentiable Adjacency

Hard binary adjacency ($A_{ij} \in \{0, 1\}$) is not differentiable with respect to atomic positions $\mathbf{r} = \{\mathbf{r}_i\}_{i=1}^n \subset \mathbb{R}^3$. We replace it with a *soft adjacency* that approximates connectivity via smooth distance-gating.

3.1 Pairwise distance

Let $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$ be the Euclidean distance between atoms i and j . Define:

$$A_{ij}(\mathbf{r}) = \sigma\left(\frac{r_c - d_{ij}}{\sigma_s}\right), \quad i \neq j, \quad A_{ii} = 0, \quad (9)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the logistic sigmoid, $r_c = 3.5 \text{ \AA}$ is the interaction cutoff (within typical bond-forming range), and $\sigma_s = 0.3 \text{ \AA}$ controls transition sharpness.

Hence $A_{ij} \approx 1$ when $d_{ij} \ll r_c$ (nearby, likely bonded), $A_{ij} \approx 0$ when $d_{ij} \gg r_c$ (distant), and a smooth interpolation near $d_{ij} = r_c$. The logistic sigmoid approximates a hard threshold while remaining everywhere differentiable. The parameter σ_s determines the width of the transition region around r_c .

3.2 Fragment-restricted masking

For scaffold inpainting the context atoms $\mathcal{C}$ are fixed; only the fragment atoms $\mathcal{F}$ can be moved. Internal context–context edges carry no information about fragment connectivity. We restrict the soft adjacency to edges involving at least one fragment atom:

$$M_{ij} = \begin{cases} 1, & i \in \mathcal{F} \text{ or } j \in \mathcal{F}, \\ 0, & \text{otherwise,} \end{cases} \quad (10)$$

$$\tilde{A}_{ij}(\mathbf{r}) = A_{ij}(\mathbf{r}) \cdot M_{ij}. \quad (11)$$

The masked soft Laplacian is then

$$\tilde{L}(\mathbf{r}) = \tilde{D}(\mathbf{r}) - \tilde{A}(\mathbf{r}), \quad \tilde{D}_{ii} = \sum_j \tilde{A}_{ij}. \quad (12)$$

Because $\tilde{A}$ and $\tilde{D}$ are smooth functions of $\mathbf{r}$, so is $\tilde{L}$.

4 Laplacian Connectivity Guidance

4.1 The Optimisation objective

We seek to maximise the Fiedler value of $\tilde{L}$ with respect to the fragment atom positions:

$$\text{maximise}_{\mathbf{r}_{\mathcal{F}}} \quad f(\mathbf{r}) = \lambda_2(\tilde{L}(\mathbf{r})). \quad (13)$$

The corresponding loss (for gradient descent) is $\mathcal{L} = -\lambda_2(\tilde{L}(\mathbf{r}))$.

4.2 Derivative of the Fiedler value

For a symmetric matrix $M(\theta)$ with simple eigenvalue λ and unit eigenvector $\mathbf{v}$ (i.e. $M\mathbf{v} = \lambda\mathbf{v}$, $\|\mathbf{v}\| = 1$), the Hellmann–Feynman theorem gives:

$$\frac{\partial \lambda}{\partial \theta} = \mathbf{v}^T \frac{\partial M}{\partial \theta} \mathbf{v}. \quad (14)$$

Proof. Differentiate $M\mathbf{v} = \lambda\mathbf{v}$ with respect to θ :

$$\frac{\partial M}{\partial \theta} \mathbf{v} + M \frac{\partial \mathbf{v}}{\partial \theta} = \frac{\partial \lambda}{\partial \theta} \mathbf{v} + \lambda \frac{\partial \mathbf{v}}{\partial \theta}. \quad (15)$$

Left-multiply by $\mathbf{v}^T$ and use $M = M^T$, so $\mathbf{v}^T M = \lambda \mathbf{v}^T$:

$$\mathbf{v}^T \frac{\partial M}{\partial \theta} \mathbf{v} + \lambda \mathbf{v}^T \frac{\partial \mathbf{v}}{\partial \theta} = \frac{\partial \lambda}{\partial \theta} \underbrace{\mathbf{v}^T \mathbf{v}}_{=1} + \lambda \mathbf{v}^T \frac{\partial \mathbf{v}}{\partial \theta}. \quad (16)$$

The $\lambda \mathbf{v}^T \partial \mathbf{v} / \partial \theta$ terms cancel on both sides (differentiating $\|\mathbf{v}\|^2 = 1$ separately confirms $\mathbf{v}^T \partial \mathbf{v} / \partial \theta = 0$), leaving $\mathbf{v}^T (\partial M / \partial \theta) \mathbf{v} = \partial \lambda / \partial \theta$, as claimed. This requires λ to be a *simple* eigenvalue, so that $\mathbf{v}$ is determined up to sign and is itself differentiable in θ ; see Section 4.4 for the degenerate case. $\square$

Applying Theorem 4.2 to $\tilde{L}(\mathbf{r})$ with eigenvalue λ_2 and eigenvector $\mathbf{v}_2$ (assumed simple; see Section 4.4):

$$\frac{\partial \lambda_2}{\partial r_{i,\alpha}} = \mathbf{v}_2^T \frac{\partial \tilde{L}}{\partial r_{i,\alpha}} \mathbf{v}_2, \quad \alpha \in \{x, y, z\}. \quad (17)$$

4.3 Gradient guidance

The per-atom force added to the denoising score is:

$$\mathbf{F}_i(\mathbf{r}) = \begin{cases} +\kappa \nabla_{\mathbf{r}_i} \lambda_2(\tilde{L}(\mathbf{r})) & \text{if } i \in \mathcal{F}, \\ \mathbf{0} & \text{if } i \in \mathcal{C}, \end{cases} \quad (18)$$

where $\kappa > 0$ is a scalar hyperparameter (`laplacian_strength`). Context atoms receive zero force since their positions are held fixed during scaffold inpainting.

4.4 Numerical stability

When the graph has multiple components, $\lambda_2 = \lambda_3 = \dots = 0$ and the eigenvalue 0 is *degenerate* (multiplicity > 1), so λ_2 is not simple and the Hellmann–Feynman derivation of Section 4.2 does not apply directly. For a degenerate eigenvalue, individual eigenvectors within the associated eigenspace are *not* in general differentiable functions of the matrix entries; only the invariant eigenspace as a whole varies smoothly [4]. The eigenvalue itself remains Lipschitz continuous but may fail to be differentiable at the degeneracy. Numerically, `torch.linalg.eigvalsh` returns *some* valid orthonormal basis of the degenerate eigenspace together with a corresponding sub-gradient, but this choice is not unique, and the resulting gradient direction can be numerically unstable near eigenvalue crossings. To handle this, any NaN or $\pm\infty$ values in the computed gradient are replaced by zero before scaling, so the force degrades (no push, rather than an arbitrary or undefined direction) when the eigenvalue structure is degenerate.

5 Properties of the Connectivity Objective

We now show that the Laplacian connectivity objective possesses several structural properties essential for scaffold inpainting. All statements below concern the masked soft Laplacian $\tilde{L}(\mathbf{r})$ and its Fiedler value $\lambda_2(\tilde{L})$.

Theorem 3 (Connectivity properties of the Laplacian objective). *The Laplacian connectivity objective satisfies the following properties:*

1. **Connectivity via the Fiedler value.** *A molecular configuration is connected if and only if $\lambda_2(\tilde{L}) > 0$, and increasing λ_2 increases the robustness of connectivity.*
2. **Monotonicity of the guidance force.** *At each denoising step, the Laplacian connectivity force moves fragment atoms in the direction of maximal instantaneous increase of $\lambda_2(\tilde{L})$.*
3. **Gradient identity.** *If $\mathbf{v}_2$ is the unit Fiedler eigenvector of $\tilde{L}$, then*

$$\nabla_{\mathbf{r}_i} \lambda_2 = \sum_j (v_{2,i} - v_{2,j})^2 \nabla_{\mathbf{r}_i} \tilde{A}_{ij}.$$

4. **Explicit sigmoid gradient.** *For $A_{ij} = \sigma(z_{ij})$ with $z_{ij} = (r_c - d_{ij})/\sigma_s$ and $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$, one has*

$$\nabla_{\mathbf{r}_i} A_{ij} = -\frac{A_{ij}(1 - A_{ij})}{\sigma_s} \hat{\mathbf{u}}_{ij}, \quad \hat{\mathbf{u}}_{ij} = \frac{\mathbf{r}_i - \mathbf{r}_j}{d_{ij}}.$$

5. **Soft-to-hard adjacency limit.** *As $\sigma_s \rightarrow 0$, the soft adjacency $\tilde{A}$ converges to the hard binary adjacency with cutoff distance r_c .*

Proof. (a) **Connectivity via the Fiedler value.** The multiplicity of the eigenvalue 0 of any graph Laplacian equals the number of connected components. Thus a connected configuration has exactly one zero eigenvalue, implying $\lambda_2 > 0$. The Courant–Fischer formula

$$\lambda_2 = \min_{\mathbf{x} \perp \mathbf{1}} \frac{\mathbf{x}^T \tilde{L} \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$$

shows that λ_2 is the minimum Dirichlet energy over all non-constant vertex assignments. Increasing λ_2 therefore increases the minimum energy required to create a near-disconnection.

(b) Monotonicity of the guidance force. The force applied to fragment atom i is

$$\mathbf{F}_i = \kappa \nabla_{\mathbf{r}_i} \lambda_2(\tilde{L}),$$

with $\kappa > 0$. Thus the update direction is precisely the gradient-ascent direction for λ_2 .

(c) Gradient identity. Since $\|\mathbf{v}_2\| = 1$, the eigenvalue admits the quadratic form

$$\lambda_2 = \frac{1}{2} \sum_{k,l} \tilde{A}_{kl} (v_{2,k} - v_{2,l})^2.$$

Differentiating with respect to $\mathbf{r}_i$ yields contributions only from terms with $k = i$ or $l = i$. Symmetry $\tilde{A}_{ij} = \tilde{A}_{ji}$ causes the two halves to combine, removing the factor $\frac{1}{2}$ and producing the stated identity.

(d) Explicit sigmoid gradient. The chain rule gives

$$\nabla_{\mathbf{r}_i} A_{ij} = \sigma'(z_{ij}) \nabla_{\mathbf{r}_i} z_{ij} = \sigma(z_{ij})(1 - \sigma(z_{ij})) \left(-\frac{1}{\sigma_s} \right) \frac{\mathbf{r}_i - \mathbf{r}_j}{d_{ij}},$$

which simplifies to the claimed expression. Since $A_{ij}(1 - A_{ij}) \geq 0$, the gradient points from $\mathbf{r}_i$ toward $\mathbf{r}_j$, with magnitude maximal at $d_{ij} = r_c$.

(e) Soft-to-hard adjacency limit. The logistic sigmoid converges to a step function as its sharpness parameter tends to zero. Thus $A_{ij} \rightarrow 1$ for $d_{ij} < r_c$ and $A_{ij} \rightarrow 0$ for $d_{ij} > r_c$, yielding the hard adjacency. Consequently, $\tilde{L}$ converges to the combinatorial Laplacian of the corresponding binary graph. $\square$

6 Integration into Reverse Diffusion

At each denoising step t the standard VE-SDE update computes a predicted noise $\boldsymbol{\epsilon}(\mathbf{r}, t)$. The Laplacian force is added as an optional correction:

$$\boldsymbol{\epsilon}_{\text{corrected}} = \boldsymbol{\epsilon}(\mathbf{r}, t) + \mathbf{F}(\mathbf{r}), \quad (19)$$

where $\mathbf{F}$ is given by equation (18). This is analogous to classifier guidance [2] but derived from a geometric connectivity objective rather than a learned classifier.

The force is off by default, and activated via the flag `--use-laplacian-connectivity` as an ablation experiment alongside the standard baseline (drift confinement only) and the valence-repulsion baseline.

7 Summary of Parameters

Symbol	Name	Default	Role
r_c	Cutoff distance	3.5 Å	Transition point of soft adjacency
σ_s	Sharpness	0.3 Å	Width of sigmoid transition
κ	Force strength	1.0	Scale of connectivity force

References

- [1] M. Fiedler, “Algebraic connectivity of graphs,” *Czechoslovak Mathematical Journal*, 23(2):298–305, 1973.
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- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2012. Chapters 6–7 (symmetric matrices and eigenvalue perturbation).